CellState Agent Scientific Report

GZMB CD8 T cell
Normal Bone Marrow (NBM) – Newly Diagnosed Multiple Myeloma (NDMM)

Execution status: completed

EXPLORATORY CONFOUNDED-DESIGN ANALYSIS: group and dataset are not independently identifiable. Features may still reflect systematic dataset effects.

Deterministic executive summary

Deterministic execution status: completed. Analysis type: combined_de_tf. Unit of inference: independent_biological_pseudobulk_replicate. EvidenceBundle e70fcedaf8e1250049a18c0560d9572dd80f8e1c4abe9705b1c517b23e3d09a5 contains 31 referenced deterministic artifacts and 5 structured warnings.

Metric	Value
CAP-DESEQ-003 · tested gene count	5213
CAP-DESEQ-003 · significant gene count	3320

Evidence critique

Confidence: **moderate**

The bundle contains extensive statistical outputs and evidence of within-bundle replication via LODO conserved features, and the design is estimable with substantial replicate counts. However, the explicit exploratory confounded-design warning (group and dataset not independently identifiable) and multiple TF-model QC warnings substantially reduce confidence in dataset-adjusted or causal claims. The results are useful for hypothesis generation and prioritization but require dataset-adjusted modeling and independent validation before stronger interpretive or translational use.

Limitations

• EXPLORATORY CONFOUNDED-DESIGN ANALYSIS: group and dataset are not independently identifiable. Features are conserved across dataset-level leave-one-out analyses but may still reflect systematic dataset effects. (see CAP-DESEQ-003:warnings.EXPLORATORY_CONFOUNDED_DESIGN)

• Deterministic evidence does not establish causality; DESeq2 outputs are association-based (see CAP-DESEQ-003:limitations).

• TF-activity results include multiple QC warnings (CAP-TF-002:warnings) including genes absent from regulons, non-estimable models, resource discordance, and some signals supported by a single resource; these weaken inference from TF models to robust regulatory conclusions.

Validated interpretation

DESeq2 pseudobulk differential expression comparing Normal Bone Marrow (NBM) to Newly Diagnosed Multiple Myeloma (NDMM) in GZMB CD8 T cells identifies widespread gene-expression differences (3,320 significant genes among 5,213 tested; 2,213 higher in NBM, 1,107 higher in NDMM) and features conserved across dataset-level leave-one-out folds. Signed TF-activity modeling produced significant and consensus TF outputs, but multiple TF-model QC warnings reduce confidence in regulator assignments. A key limitation is that group and dataset are not independently identifiable in this exploratory LODO analysis, so some signals may reflect dataset effects rather than true biology. Results are appropriate for hypothesis generation and prioritization but require dataset-adjusted modeling and independent/orthogonal validation before stronger regulatory or causal claims.

Observations

• Analysis context: cell state GZMB CD8 T cell; contrast Normal Bone Marrow (NBM) – Newly Diagnosed Multiple Myeloma (NDMM); inference_class exploratory_lodo_conserved (CAP-DESEQ-003, CAP-TF-002).

• Dataset footprint and design: 167,355 contributing cells; eligible biological pseudobulk replicates NBM=55, NDMM=233; genes in atlas=15,412; minimum cells per replicate reported 100 (deterministic_evidence.atlas_qc).

• Design and estimability: DESeq2 design formula ~ group (group coefficient 'NBM - NDMM'), design full-rank and estimable, residual degrees of freedom = 286 (CAP-DESEQ-003:design_assessment).

• Differential expression outputs (DESeq2, CAP-DESEQ-003): tested_gene_count = 5,213 (retained genes 5,213); significant_gene_count = 3,320; of significant genes, 2,213 are higher in NBM and 1,107 are higher in NDMM (CAP-DESEQ-003:deseq2_results, deterministic_evidence.CAP-DESEQ-003).

• Reproducibility within the bundle: features conserved across dataset-level leave-one-out (LODO) analyses (CAP-DESEQ-003:conserved_features and CAP-DESEQ-003:exploratory_lodo_summary), indicating LODO-level conservation but not dataset-adjusted inference.

• TF-activity modeling (CAP-TF-002) completed with signed regulon resources and produced per-resource TF activity outputs, a significant TF activity table, and a consensus TF activity table (CAP-TF-002:tf_activity_by_resource, CAP-TF-002:significant_tf_activity, CAP-TF-002:tf_activity_consensus).

• TF-activity QC warnings (CAP-TF-002 warnings): 1,373 eligible genes absent from all signed regulons retained in the model universe; 944 resource-program-TF models non-estimable; 264 resource discordance instances preserved as QC; 12 significant activities supported by only one resource (CAP-TF-002 warnings).

• Statistical execution: DESeq2 pseudobulk differential expression was performed and full unadjusted statistics are available (CAP-DESEQ-003:full_unadjusted_deseq2_results).

• Status and scope: both CAP-DESEQ-003 and CAP-TF-002 completed with warnings; outputs include artifacts suitable for per-gene and per-dataset inspection (artifact list in evidence bundle).

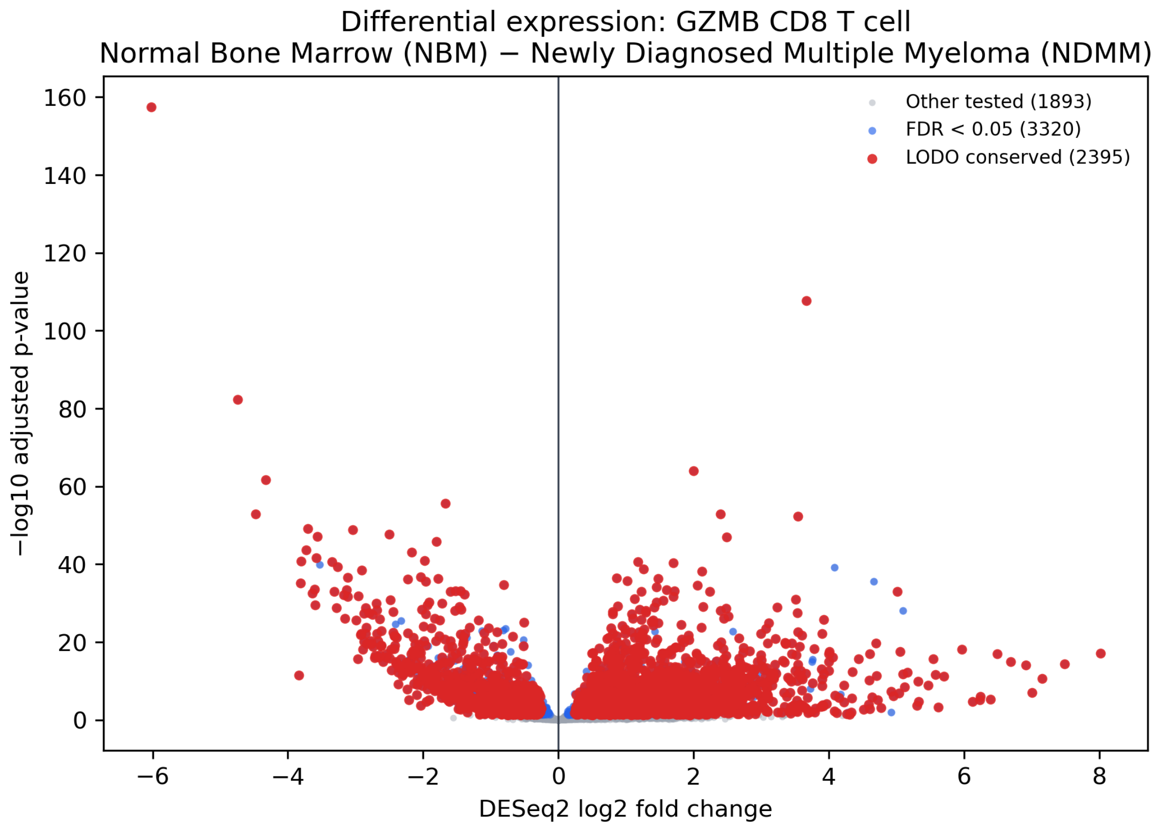
Explicit hypotheses

• Hypothesis: Some fraction of the 3,320 genes differentially expressed between NBM and NDMM in GZMB CD8 T cells reflect biologically reproducible differences across datasets, as indicated by LODO-conserved features (CAP-DESEQ-003:conserved_features).

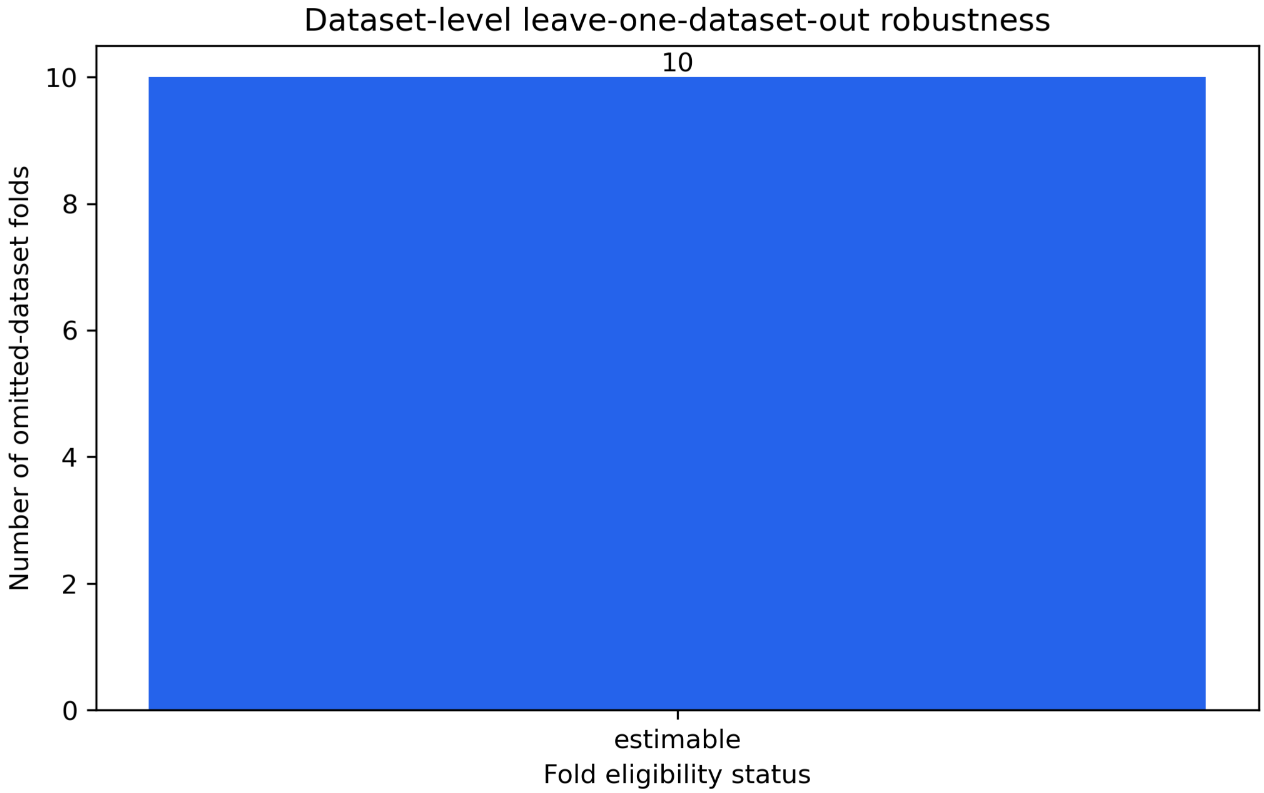
• Hypothesis: A subset of TFs identified in CAP-TF-002 consensus outputs may contribute to the observed expression differences; however, this is model-based and weakened by TF-activity QC warnings (CAP-TF-002:warnings).

• Hypothesis: Some observed differential-expression signals may be driven by dataset-level effects rather than underlying biology because group and dataset are not independently identifiable in this exploratory analysis (CAP-DESEQ-003:design_assessment warnings EXPLORATORY_CONFOUNDED_DESIGN).

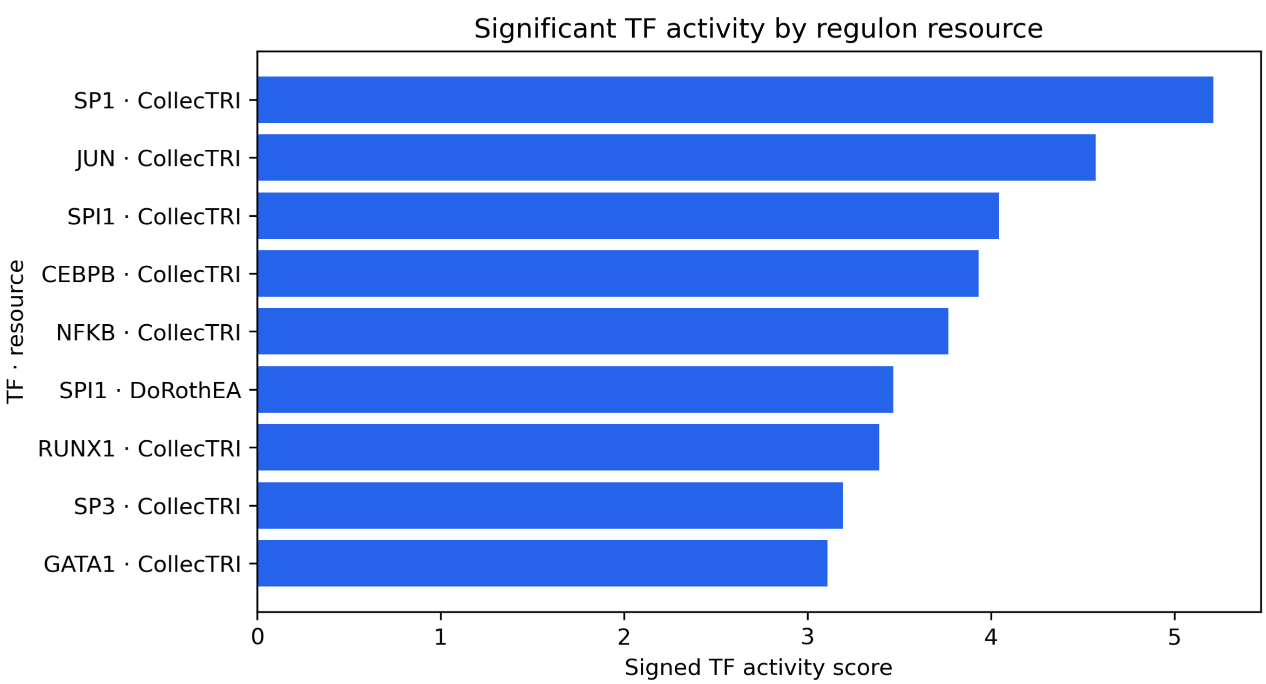
Figures



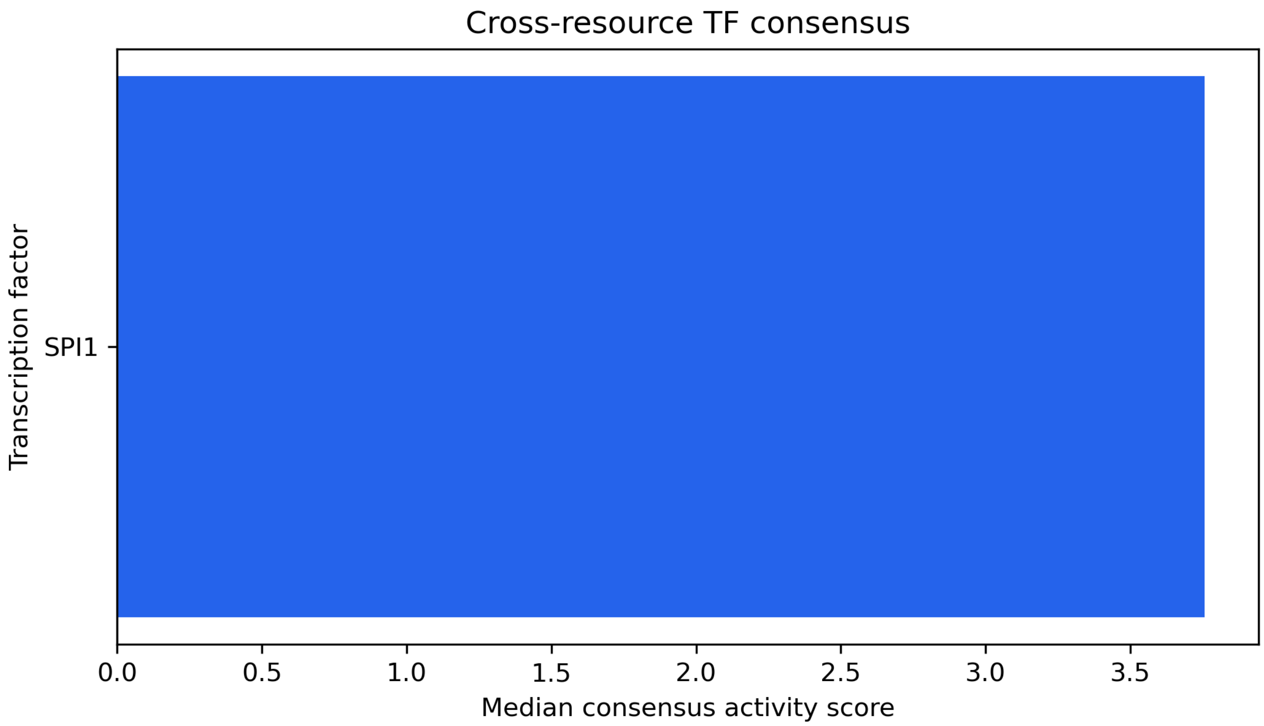
DESeq2 effect size versus multiple-testing-adjusted significance; positive values follow the configured group A minus group B contrast. Exploratory confounded design: group and dataset are not independently identifiable.



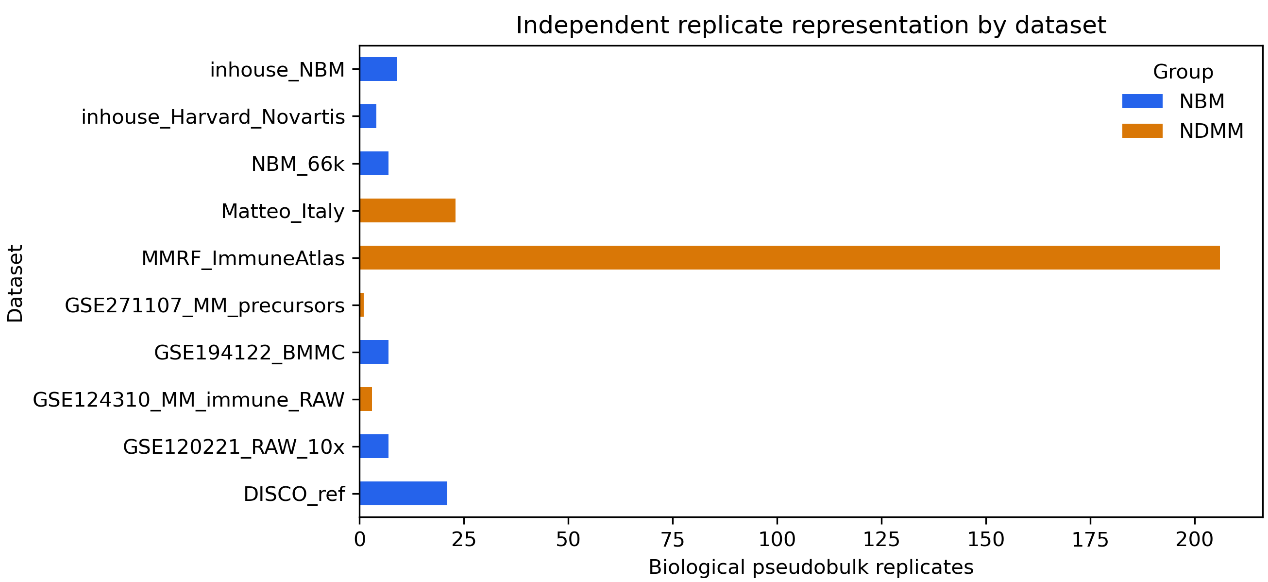
Eligibility of 10 dataset-level LODO folds; estimable folds contribute to feature stability summaries. Exploratory confounded design: group and dataset are not independently identifiable.



Top absolute significant TF activity estimates within each resource (9 displayed); sign follows the configured contrast orientation.



Directionally concordant cross-resource TF estimates (1 displayed); positive and negative scores follow the configured contrast.



Distribution of 288 independent biological pseudobulk replicates across 10 datasets and 2 comparison groups. Exploratory confounded design: group and dataset are not independently identifiable.